

INVESTIGATION OF SUSPECTED SOURCE CODE EXFILTRATION

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Digital Forensics - Project 3



INCIDENT OVERVIEW

- Small software company reported suspected source code theft
- Company management requested a forensic investigation to reconstruct the suspect's activities
- Windows file server IP address –192.168.4.5
- Robert's assigned workstation IP –192.168.5.5
- Evidence provided –
 - Packet capture files (pcap file1 and file2)
 - Network traffic evidence
 - SMB communication records

INVESTIGATION OBJECTIVES

- Analyze packet captures for suspicious network activity
 - Examine SMB communication using Wireshark
- Identify authenticated file-share access activity
- Determine evidence of –
 - Shared folder access
 - External outbound communication
 - Potential data exfiltration behavior
- Reconstruct a timeline of the incident using timestamps
- Maintain forensic integrity and proper evidence handling

TOOLS USED

- **Linux/Kali Tools** - Evidence examination and forensic analysis
- **Wireshark** - Packet capture and network traffic analysis
- **Microsoft Excel** - Timeline reconstruction and examiner notes
- **Microsoft PowerPoint** - Investigation reporting and presentation

DIGITAL FORENSIC PROCESS

- Identification Incident and suspicious activity
- Preservation Evidence preserved to maintain integrity
- Collection Packet capture files collected for analysis
- Examination SMB traffic and external communication
- Analysis Timeline reconstruction and event correlation
- Reporting Findings

WIRESHARK ANALYSIS DEMO

ATTACK ANALYSIS SUMMARY

Evidence Source	Activity Identified	Adversary Goal
file1.pcap	SMB2 authentication and shared folder access activity identified	Access internal shared resources
file1.pcap	Tree Connect Request to shared folder observed	Establish SMB-based file access
file2.pcap	TLS encrypted outbound communication detected	Transfer data externally
file2.pcap	Communication with external IP 20.42.65.91 over port 443	Maintain outbound connection
file2.pcap	Multiple encrypted application data packets transmitted	Possible source code exfiltration
Recovered SMB Object	Source code file login_module_v1.c successfully recovered	Access sensitive source code

ATTACK TIMELINE

Time 2026-05-05	Activity	Evidence Source
21:37	SMB2 negotiation and session setup initiated	file1.pcap
21:37	Authenticated access to shared resources confirmed	file1.pcap
21:37	Tree Connect Request established to shared folder	file1.pcap
21:44	TLSv1.2 communication initiated with external host	file2.pcap
21:44	Multiple encrypted application data transfers observed	file2.pcap
21:44	Outbound communication session terminated	file2.pcap
21:44	Source code file login_module_v1.c recovered from SMB export	file2.pcap

KEY FINDINGS

- SMB file-sharing traffic identified
- Authenticated access to shared resources confirmed
- Shared folder connections established
- Robert's assigned workstation initiated outbound encrypted communications
- Outbound traffic to external IP 20.42.65.91 observed
- Activity consistent with possible source code exfiltration behavior

CONCLUSION

- Internal SMB-based file transfer activity was confirmed
- Sensitive source code file was successfully recovered
- Robert's assigned workstation generated outbound encrypted communications
- Network activity is consistent with possible source code exfiltration behavior
- TLS encryption prevented direct verification of transferred content
- Evidence supports attribution to Robert's assigned workstation

RECOMMENDATIONS

- Implement centralized logging and endpoint monitoring
- Enforce least-privilege access for shared resources
- Monitor outbound encrypted traffic for suspicious activity
- Deploy Data Loss Prevention (DLP) technologies
- Enable alerting for unauthorized SMB file transfers
- Conduct regular security audits and forensic reviews

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THANK YOU!